

Capstone Proposal

Methoda and Data

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Data Processing and Analysis

Software and Packages

All data processing, analysis, and modeling will be conducted using free open source software and packages to maximize accessibility and reproducibility. The majority of data manipulation and processing will be performed in the R programming language using the tidyverse suite of packages (Wickham et al. 2019; R Core Team 2024). Any data operations that need to be “spatially aware” will be performed using the Simple Features package (Pebesma 2018).

Analysis will be scripted and ran in Positron, an open-source integrated development environment developed by Posit that supports R, Python, scripts, notebooks, integrated consoles, along with interactive plots and data viewers.

Study Area

The study area for this project is the Pensacola Bay System which includes portions of Pensacola Bay, Escambia Bay, East Bay and Blackwater Bay. Boundaries of this study area were achieved by accessing Florida Department of Protections (FDEP) Waterbod IDs feature layer hosted on ArcGIS Online (FDEP 2005). ArcGIS allows direct access to hosted feature services via several formats. To avoid any potential compatibility issues, the GeoJSON format was chosen as this format is non-proprietary. Once retrieved, the dataset was filtered to include only relevant WBIDs (WBID = 548) and then combined to create a singular study area polygon. Creating the study area polygon is an important first step as it will be needed to extract relevant environmental data from much larger datasets in the coming data gathering.

```
# Create study area using FDEP WBID feature service

json_url <- "https://ca.dep.state.fl.us/arcgis/rest/services/OpenData/WBIDS/MapServer/0/query"
wbid <- read_sf(json_url)

# List of relevant WBIDs
study_area_wbid_list <- c(
  "548GA",
  "548GB",
  "548H",
  "548C",
  "548D",
  "548B",
  "548AA",
  "548AB",
  "548AC"
)

study_area <- wbid %>%
  filter(WBID %in% study_area_wbid_list) %>% # Filter results to only include desired WBIDs
  summarise() # dissolve all polygons into one singular geometry
```

References

- FDEP. 2005. "Waterbody IDs (WBIDs)." Geospatial Open Data; Florida Department of Environmental Protection. <https://geodata.dep.state.fl.us/datasets/FDEP::waterbody-ids-wbids/about>.
- Pebesma, Edzer. 2018. "Simple Features for R: Standardized Support for Spatial Vector Data." *The R Journal* 10 (1): 439–46. <https://doi.org/10.32614/RJ-2018-009>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.